Priya Goyal

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Permanent U.S. Resident
~71K citations

ABOUT ME

I am a technical leader specializing in building the research and engineering foundations around increasingly capable Agentic AI systems: evaluation, synthetic data, reward signals, and feedback loops that turn model capabilities into reliable systems and scaling these foundational solutions across organizations.

EMPLOYMENT

Sr. Staff Research Engineer, LinkedIn CoreAI — Dec 2024 - Present, New York, NY

- Founded LinkedIn Agentic Search, scaling the initiative from concept to a company-priority and ~50+ person effort that defined LinkedIn's agentic AI future.
- Architected a novel Agentic Evaluation framework from scratch for a subjective problem in a cold start 0->1 setup: defined metrics, rubrics, agent-as-judge, user feedback hill climbing loop, human annotation, grounded / real synthetic data generation, multi-turn eval. The eval framework is used in every corner of Agentic Search effort and also enables all model training (RL, SFT) and critical launch-readiness assessments for Enterprise and consumer products.
- Also leading a broad effort on recursive self-improvement
- Appointed by executive leadership (CTO and Chief AI Officer) to lead NextGen company-wide AI Evaluation, authoring the standardized AI Eval playbook and driving adoption across multiple product pillars including Hiring Assistant, Feed, News, Sales Navigation Agent, Career Agents.
- Served as a top contributor (largest #PRs, ~500 PR reviews) while coordinating technical alignment and daily execution rhythms across a cross-functional team of 50+ ICs.

Datology AI, Founding Member of Technical Staff — Jan 2024 - Oct 2024, New York, NY
Founding Researcher

- Led the research on how to curate high quality data such that it leads to on-par / better accuracy but with much better efficiency (e.g. in 50% of the training time). This also involved figuring out a non-trivial problem of how to evaluate and compare the quality of two different datasets.
- Built the company infrastructure to support data quality research including thousands of experiments, evaluation.
- Senior most researcher in the company informing clear research plans and priorities for the company including infrastructure investments.
- Also led customer collaborations on data curation leading to more efficient and higher quality language models for the customer.

Google Deepmind — Aug 2022 – Jan 2024, New York, NY
Staff Research Engineer

- Started and led a moonshot project around multimodal personalization with the goal of uplifting LLMs such as Gemini to add personalized value to daily user life by understanding users preferences derived from Google photos augmented with memory. Conducted the open-ended research on fine-grained segmentation in images to enable the personalization.
- Also worked on understanding the biases and mitigation strategies in multimodal models like the CLIP (ICLR'24) paper.

Facebook AI Research — Oct 2016 – Aug 2022, New York, NY
Staff Research Engineer

- Tech Lead projects on self-supervised learning on a large internet scale noisy, unbounded/uncurated data, producing high quality models with better quality features compared to weakly supervised / supervised learning. Paper published with extensive media coverage (links below).
- Also published works on thoroughly evaluating many self-supervised approaches which led to clear identification of the gap needed to be addressed with foundational research to unleash benefits of self-supervision in computer vision.
- Also designed and open sourced the fairness benchmarks for computer vision models (published in FAccT 2022).

- Also led the engineering work on a simple one-stage Object Detector called RetinaNet and novel Focal Loss (ICCV'17 best paper).
- Extensive research experience in computer vision and deep learning – ~71K+ plus citations, won best paper award at ICCV, led and released open-source self-supervision libraries (such as VISSL) and organized a self-supervised learning challenge at ICCV workshop.
- Large scale vision modeling - built largest (to date) dense computer vision models (10 billion parameters) and first author of foundational distributed training work – ImageNet in 1hour which sparked an industry-wide race of training faster than 1 hour (Google, Intel, IBM, etc.)
- Extensive Media Coverage of many projects highlighted below.

Facebook Jan 2016 – Sept 2016, *London, UK*

Software Engineer

- Built deep learning models to improve the knowledge and entities graph of Facebook. Led the work on building neural networks from highly noisy data.

EDUCATION

Indian Institute of Technology (IIT) Kanpur India, 2010 - 2015

Bachelor and Masters, Mathematics and Scientific Computing

M.Sc. Advisor: Prof. Harish Karnick

Thesis Title: Study of auto-encodings in the hidden units of deep neural networks.

GPA: 9.1/10 (Ranked 2 of 50 in graduating class).

RESEARCH INTERESTS

Agentic Eval: Agent-as-Judge and Fine-grained dynamic objective rubrics for subjective problems (consumer / enterprise).

Synthetic Agentic Data: High quality grounded synthetic data generation for agentic products.

Multimodal learning: learning with many modalities including text, images.

Computer Vision / Machine Learning: learning with limited supervision (from uncurated data).

Socially Responsible AI: diagnosing and mitigating harms, model privacy and robustness.

Large scale vision models: pushing the limits in terms of data and model scale.

PUBLICATIONS

- *Clip the bias: How useful is balancing data in multimodal learnings?* - Ibrahim Alabdulmohsin, Xiao Wang, Andreas Peter Steiner, Priya Goyal, Alexander D'Amour, Xiaohua Zhai, **ICLR'24**
- *Vision Models Are More Robust And Fair When Pretrained On Uncurated Images Without Supervision* - Priya Goyal, Quentin Duval, Isaac Seessel, Mathilde Caron, Ishan Misra, Levent Sagun, Armand Joulin, Piotr Bojanowski, **arXiv 2022**
- *Fairness Indicators for Systematic Assessments of Visual Feature Extractors* - Priya Goyal, Adriana Romero Soriano, Caner Hazirbas, Levent Sagun, Nicolas Usunier, **FAccT, 2022**
- *A Self-Supervised Descriptor for Image Copy Detection* - Ed Pizzi, Sreya Dutta Roy, Sugosh Nagavara Ravindra, Priya Goyal, Matthijs Douze, **CVPR 2022**
- *Fully Sharded Data Parallel: faster AI training with fewer GPUs* - Myle Ott, Sam Shleifer, Min Xu, Priya Goyal, Quentin Duval, Vittorio Caggiano, **Facebook Engineering blog, 2021**
- *VISSL: A library for state-of-the-art self-supervised learning from images* - Priya Goyal, Quentin Duval, Jeremy Reizenstein, Matthew Leavitt, Min Xu, Benjamin Lefaudeux, Mannat Singh, Vinicius Reis, Mathilde Caron, Piotr Bojanowski, Armand Joulin, Ishan Misra, **2021**
- *Self-supervised pretraining of visual features in the wild* - Priya Goyal, Mathilde Caron, Benjamin Lefaudeux, Min Xu, Pengchao Wang, Vivek Pai, Mannat Singh, Vitaliy Liptchinsky, Ishan Misra, Armand Joulin, Piotr Bojanowski, **arXiv 2021**
- *Unsupervised Learning of Visual Features by Contrasting Cluster Assignments* - Mathilde Caron, Ishan Misra, Julien Mairal, Priya Goyal, Piotr Bojanowski, Armand Joulin, **NeurIPS 2020**
- *Scaling and Benchmarking Self-Supervised Visual Representation Learning* - Priya Goyal, Dhruv Mahajan, Abhinav Gupta*, Ishan Misra*, **ICCV 2019**
- *The next 700 accelerated layers: From mathematical expressions of network computation graphs to accelerated gpu kernels, automatically* - Nicolas Vasilache, Oleksandr Zinenko, Theodoros Theodoridis, Priya Goyal, Zachary Devito, William S Moses, Sven Verdoolaege, Andrew Adams, Albert Cohen, **ACM TACO 2019**
- *Focal Loss for Dense Object Detection* - Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, Piotr Dollár, **ICCV 2017 (*best student paper award*)**
- *Accurate, Large Minibatch SGD: Training ImageNet in 1 Hour* - Priya Goyal, Piotr Dollár, Ross Girshick, Pieter Noordhuis, Lukasz Wesolowski, Aapo Kyrola, Andrew Tulloch, Yangqing Jia, Kaiming He, **arXiv 2017**

Tech Reports

o Peer-to-peer insult detection in online communities - Priya Goyal, Gaganpreet Singh Kalra, 2013

WORKSHOP CHALLENGE

Facebook AI Self-Supervision Learning Challenge ICCV, 2019

Extreme Computer Vision Workshop *Seoul, Korea*

Organized the first ever self-supervised challenge aimed at benchmarking representation learning on a variety of tasks. Wrote the transfer learning code and setup the challenge evaluation/submission server.

INVITED TALKS

- o SEER, a self-supervised billion parameter computer vision model - Facebook AI Innovation Summit, 2021
- o Scaling and Benchmarking Self-Supervised learning – Carnegie Mellon University, April 2019.
- o Facebook North America Women in AI Leadership Summit, 2019 – 10K attendees.
- o ImageNet-1k training in 1 hour – Caffe2 Meetup, Hawaii, CVPR 2017
- o ImageNet in 1 hour – Deep Learning at Supercomputer Scale, NeurIPS 2017

MEDIA COVERAGE

- TechCrunch article on ImageNet in 1-Hour.
- CNBC article on SEER (training A.I. to "see").
- NVIDIA Developer on ImageNet on 1-Hour.
- Geekwire on ImageNet in 1-Hour.
- NVIDIA Developer on Self-supervised learning beating SOTA Computer vision models.
- WIRED article on AI Teaching Itself to See With Less Human Help.
- CNET on training computers to learn like humans do.
- ImageNet in 1-Hour at NeurIPS 2017 Supercomputing workshop.

AWARDS

- o Best Student Paper Award, ICCV 2017.
- o Facebook Grace Hopper Scholarship, 2014 and Google Global Scholarship, 2013.
- o Academic Excellence Award 2011-12 and 2012-13 – given to the top 5% of 850 students.
- o Best Poster Award (amongst 60 posters) presented in SURGE, IIT Kanpur 2012.
- o SURGE undergraduate research fellowship (2012), Indian Institute of Technology (IIT), Kanpur, India.
- o National Talent Search Scholarship, 2008 – given to top 0.01% high school students all over India.

ACADEMIC SERVICE

Conference reviewer: Computer Vision and Pattern Recognition (2018 – 2022), International Conference on Computer Vision (2018 – 2022), European Conference on Computer Vision (2018 – 2022), International Conference on Learned Representations (2018 – 2022), International Conference on Machine Learning (2018 – 2022), Neural Information Processing Systems (2018 – 2022)

INTERNSHIPS

Facebook May-August 2015

Advisor: Kay Rottmann, Juan Miguel Pino, Language Technology Team *Menlo Park, CA, USA*

Built profanity detection classifier to detect profanity in the machine translations produced by Facebook. Built classifiers for all the languages and automated the re-training of any classifier.

Goldman Sachs May–July 2014

Advisor: Andrei Bergners, Panna Pavangadkar, Developer Practices Group *New York, US*

Built software for making the firm approved SDLC build cloud dynamic and self-aware. The product analyzes nearly 3000 product builds across the firm on a daily basis.

Yahoo! May–July 2013

Advisor: Bennet Manuel, Yahoo! Right Media Exchange *Bangalore, India*

Built visualization tool for tracking the usage of web bugs in online display ads to detect malicious vendors - helped increase the revenue of the company by curbing ban on minor defaulters.

Carnegie Mellon University Winter School Dec 2012–Jan 2013

Advisor: Prof. Carolyn P. Rosé, Department of Computer Science, CMU

Develop an interactive tutor moderated chat system for automated analysis and dynamic support for an online collaborative learning process. Proposed and implemented 2-pass architecture model for modelling the students' conversation and building an agent prototype capable of detecting distraction/ confusion among students and provide necessary support to ensure effective learning.

Summer Undergraduate Research Grant for Excellence May-July 2012

Advisor: Prof. Akash Anan, Department of Mathematics and Statistics *IIT Kanpur, India*

Developed efficient algorithms for projecting the point cloud on the projection surface. Automated the process of 3-D surface generation.